



S⁴D: Style and Structure Similarity for Site Domains

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Περίληψη

Τα τελευταία χρόνια, οι ιστότοποι έχουν γίνει πιο πολύπλοκοι και ποικιλόμορφοι ως προς τον σχεδιασμό και τη δομή τους. Παρά την ποικιλομορφία αυτή, οι ιστότοποι που βρίσκονται στον ίδιο θεματικό τομέα ή εξυπηρετούν παρόμοιους σκοπούς συχνά μοιράζονται κοινά παρουσιαστικά και δομικά μοτίβα. Η αναγνώριση και η σύγκριση αυτών των μοτίβων σε διαφορετικούς τομείς ιστότοπων είναι πολύτιμη για την επίτευξη αποτελεσμάτων ως προς την ταξινόμηση ιστοσελίδων, την αυτοματοποιημένη ανάλυση διεπαφών χρήστη (UIs) και την αποτύπωση των θεματικών τους τομέων. Ωστόσο, η δομή και η οπτική παρουσίαση των περιεχομένων ιστού είναι συχνά μη τυποποιημένη και ασυνεπής, αυξάνοντας τη δυσκολία τέτοιων αναλύσεων. Σε αυτήν την πτυχιακή εργασία, παρουσιάζεται το **S⁴D**, ένα σύστημα που αυτοματοποιεί τη διαδικασία εξαγωγής, μοντελοποίησης και σύγκρισης των παρουσιαστικών και δομικών ομοιοτήτων διαφορετικών ιστοτόπων. Το σύστημα αναλύει ένα σύνολο ιστοσελίδων μέσω τεχνικών αυτοματοποιημένης περιήγησης, μοντελοποιώντας τα CSS χαρακτηριστικά των στοιχείων τους και οργανώνοντάς τα με τρόπο που επιτρέπει την περαιτέρω ανάλυση. Συγκεκριμένα, μετατρέποντας τα στοιχεία σελίδας σε διανυσματικές αναπαραστάσεις και εφαρμόζοντας τεχνικές μη-επιβλεπόμενης συσταδοποίησης, το σύστημα εντοπίζει ουσιαστικές συστάδες των στοιχείων βάσει των οπτικών τους χαρακτηριστικών. Χρησιμοποιώντας αυτήν την συσταδοποιημένη αναπαράσταση, το σύστημα μοντελοποιεί τα παραγόμενα δεδομένα για κάθε ιστότοπο, εξάγει μοτίβα διάταξης στοιχείων και υπολογίζει την ομοιότητα μεταξύ τους χρησιμοποιώντας διαφορετικές μετρικές απόστασης. Τέλος, το σύστημα παρουσιάζει έναν πίνακα ομοιότητας μεταξύ των ιστοτόπων, προσφέροντας πληροφορίες σχετικά με τη δομική και παρουσιαστική ομοιότητα των ιστοσελίδων τους.

Abstract

In recent years, websites have become increasingly complex and diverse in design and structure. Despite this diversity, websites belonging to the same domain or serving similar purposes frequently exhibit similar stylistic and structural characteristics. Identifying and comparing these patterns across different site domains is valuable for tasks such as web classification, automated user interface (UI) analysis, and domain topic labeling. However, the structure and visual presentation of web content is often unstandardized and inconsistent, making such analysis a challenging task. This thesis introduces **S⁴D**, a system that automates the process of extracting, modeling, and comparing the style and structure of websites across different domains. The system parses a set of webpages using browser automation techniques, modeling CSS-related properties of web elements, and structuring them in a way that enables further analysis. By transforming page elements into vector representations and applying unsupervised clustering techniques, the system identifies meaningful groupings of visual and structural components. Using this clustered representation, the system exploits the resulting data modeling of each domain, extracting high-level patterns of element arrangement, and computes the inter-domain similarity using multiple distance metrics. Finally, the system presents a similarity matrix between domains, providing insight into the structural and stylistic resemblance of their webpages.

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Chapter 1

Introduction

This thesis presents **S⁴D**, a system for analyzing and detecting similarities between websites. **S⁴D** is a multi-layered system that aims to facilitate the modeling and analysis of websites' visual and structural similarity. Through a variety of functionalities and a Graphical User Interface (GUI), it is able to yield meaningful insight into the websites' visual and structural similarity analysis process, ultimately generating a similarity matrix for all input webpages.

1.1 Problem Statement

The current landscape of web analysis tools focuses primarily on content-based analysis, often overlooking the significance of visual design patterns and structural similarities between websites. In the era of rapidly expanding web content, understanding the structural similarity between websites has become increasingly important for various applications including web mining, content analysis, and automated web processing. However, traditional approaches face several critical challenges.

1.1.1 Web Structure Analysis Challenges

Modern web structure analysis presents several unique technical challenges that our S⁴D system specifically addresses:

- **DOM Complexity and Recursive Processing.** Modern websites contain deeply nested DOM structures with thousands of elements, requiring recursive traversal algorithms to accurately capture hierarchical relationships. The sheer depth and breadth of contemporary HTML trees make consistent and scalable structural analysis a non-trivial engineering challenge.
- **CSS Style Diversity and Standardization.** The variety of CSS styling approaches across websites creates inconsistency in structural representation, making cross-site comparisons particularly challenging. Different sites express the same visual properties through differing units, formats, and conventions, requiring unified normalization before any meaningful comparison can take place.
- **Attribute Extraction and Feature Engineering.** Extracting meaningful features from HTML elements involves processing hundreds of CSS properties per element. Identifying which properties carry structural significance — and how to normalize them for comparison — is central to building a reliable similarity model.
- **Scalability and Performance Optimization.** Processing large numbers of websites demands efficient algorithms and robust memory management. As the number of analyzed domains grows,

the computational overhead of feature extraction and comparison increases substantially, requiring careful architectural choices to ensure reliable performance at scale.

- **Multi-dimensional Similarity Computation.** Defining meaningful similarity measures for web structures involves multiple algorithmic challenges, since web pages differ across visual, structural, and positional dimensions simultaneously. No single metric fully captures the richness of inter-site similarity, making combined approaches necessary.
- **Clustering Parameter Optimization.** Achieving optimal clustering results requires careful parameter tuning, as different web datasets exhibit varying densities and structural patterns. Selecting appropriate parameters without overfitting to a specific domain is a non-trivial problem that demands principled search strategies.
- **Cross-Domain Pattern Recognition.** Identifying design trends across different domains requires sophisticated comparison frameworks capable of abstracting away domain-specific content while preserving structural and stylistic signals. This becomes especially challenging when the overall design language varies significantly between domains.

1.1.2 Research Motivation

The need for automated web structural analysis stems from several key factors:

- **Content Management:** Organizations need tools to understand and manage large collections of web content based on structural and visual characteristics.
- **Design Pattern Analysis:** Web developers and designers benefit from understanding common structural patterns and visual trends across websites in their domain.
- **Accessibility Evaluation:** Automated structural analysis can help identify accessibility issues and design inconsistencies across multiple web properties.
- **Web Analytics Enhancement:** Structural similarity analysis provides deeper insights into web-site categorization and user experience patterns beyond traditional content-based metrics.

1.2 Our Methodology and Contributions

S⁴D addresses the challenges outlined above through a cohesive set of methods and components. To handle DOM complexity, the system uses unique ID generation and hierarchical tree processing to manage parent-child relationships across depth levels 1–10 while maintaining structural integrity throughout analysis. For CSS diversity, sophisticated parsing and normalization pipelines process pixel values (**px**), percentages (%), and relative units in a unified manner, enabling consistent cross-site comparisons.

Attribute extraction is guided by selective filtering focused on structurally significant properties such as **font-size**, **text-align**, **line-height**, and **cursor**, with dedicated handling for color space conversions and unit standardization. Scalability is achieved through domain-based processing modes, ChromeDriver optimization, and parallel data processing, enabling reliable batch analysis across large website collections.

For similarity computation, the system combines HDBSCAN clustering with DBCV (Density-Based Cluster Validation) scoring [18], cosine similarity matrices, and role vector analysis to jointly capture structural, visual, and positional relationships between elements. Clustering parameters are tuned via grid search over cluster size ranges and selection epsilon values, with automatic outlier reassignment to the

nearest cluster centroid using Euclidean distance. Cross-domain analysis is enabled through domain-level aggregation, transition matrices between design elements, and the use of the cosine similarity metric.

This thesis makes the following key contributions:

- **Novel feature extraction methodology:** Development of comprehensive techniques for extracting both structural and visual features from websites
- **Advanced similarity metrics:** Introduction of new metrics specifically designed for website comparison
- **Integrated clustering approach:** Implementation of clustering algorithms optimized for website analysis
- **Comprehensive evaluation framework:** Establishment of evaluation metrics and datasets for assessing website similarity systems
- **Practical applications:** Demonstration of real-world applications in web design analysis and automated categorization

1.3 Research Objectives

To tackle the challenges raised above, the primary objectives of this research are:

1. To develop a robust system for extracting and analyzing structural and visual features from websites
2. To create effective similarity metrics that can accurately measure relationships between different websites
3. To implement clustering algorithms that can group websites based on their design and structural characteristics
4. To provide a comprehensive comparison framework for website analysis across different domains
5. To validate the effectiveness of the proposed system through extensive testing and evaluation

1.4 Thesis Structure

The remainder of this thesis is organized as follows:

- **Chapter 2: Literature Review** presents a comprehensive review of existing approaches to website analysis, similarity measurement, and clustering techniques relevant to web data.
- **Chapter 3: Methodology** describes the proposed S⁴D system architecture, including feature extraction methods, similarity metrics, and clustering algorithms.
- **Chapter 4: Implementation** details the technical implementation of the system, including data structures, algorithms, and software architecture decisions.
- **Chapter 5: Experimental Design** outlines the experimental methodology, datasets used, and evaluation metrics employed to assess the system's performance.
- **Chapter 6: Results and Analysis** presents the experimental results, performance analysis, and comparison with existing approaches.

- **Chapter 7: Discussion** provides an in-depth discussion of the findings, limitations, and implications of the research.
- **Chapter 8: Conclusion and Future Work** summarizes the key findings, contributions, and suggests directions for future research.

Chapter 2

Related Work

This chapter reviews research related to the structural and visual analysis of web pages, highlighting prior work in web parsing, information extraction, representation learning, DOM-based similarity, main content extraction, and density-based clustering. The discussion emphasizes how these approaches inform, contrast with, and complement the S⁴D system. The review is organized into five major areas: (1) structural understanding and representation of web documents, (2) DOM element similarity and tree-based comparison, (3) web template and design pattern detection, (4) web content extraction and main content identification, and (5) density-based clustering and its validation. A comparative discussion and identification of research gaps close the chapter.

2.1 Structural Understanding and Representation of Web Documents

Understanding the internal structure of web pages has been a central theme in web analysis research, with approaches ranging from dataset construction to deep learning representations and clustering-based organization.

One of the earliest challenges in web analysis has been the ability to combine textual semantics with structural information. Chen et al. [4] introduced **WebSRC**, a large-scale dataset for structural reading comprehension comprising 400K question–answer pairs across 6.4K web pages, combining HTML source code, screenshots, and metadata. Their work demonstrates that effective web understanding requires not only semantic interpretation but also the hierarchical organization of HTML elements. While WebSRC focuses on question answering, its emphasis on DOM-aware structural comprehension is directly relevant to S⁴D’s approach of exploiting the hierarchy of DOM nodes as a primary signal for layout analysis. Unlike WebSRC, however, S⁴D does not address query answering but targets structural and stylistic similarity across entire domains.

Building on this idea of structural representation, Deng et al. [6] proposed **DOM-LM**, a transformer-based model that jointly encodes both textual content and DOM structure to learn generalizable representations of web pages. DOM-LM addresses the shortcomings of methods that rely solely on visual rendering or plain-text representations, achieving strong generalization in few-shot and zero-shot tasks such as attribute extraction and information retrieval. In contrast, S⁴D applies structural encoding not for semantic extraction but to identify recurring patterns of element arrangement across domains. Rather than fine-tuned embeddings, S⁴D operates on raw CSS feature vectors derived directly from DOM elements, enabling an unsupervised, label-free pipeline.

Hambley et al. [11] approached the problem of structural web representation from a different an-

gle. In their work on **web structure-derived clustering for optimised accessibility evaluation**, they empirically compared five representations—tag sequences, tree structure, full DOM tree encodings, content-based, and combined structure-content—and evaluated their suitability for clustering large web sites. Their central finding is that DOM structural features (tag sequences, tree topology) produce higher-quality and more coherent clusters than content-based alternatives; furthermore, incorporating structural components significantly improves the representativeness of the resulting sample when auditing accessibility. This directly validates S⁴D’s architectural choice of working at the level of structural CSS features rather than page text. Hambley et al. apply clustering to reduce thousands of pages to archetypal representatives for accessibility audit on sites of up to 500 pages, whereas S⁴D scales clustering to the element level and aggregates results across domains.

Huynh et al. [12] introduced a DOM-structural cohesion analysis approach to **web page segmentation**, treating element types, attributes, and structural cohesion as primary features. Their method achieves 64% F_{B3} on a benchmark of 1,969 web pages, surpassing prior state-of-the-art segmentation methods, without relying on screenshot-based rendering. Their work confirms that element-level DOM attributes are strong enough signals to segment and partition page structure in a purely structural pipeline—the same philosophy underlying S⁴D’s CSS feature extraction stage. The key difference is scope: Huynh et al. produce intra-page segmentation boundaries, whereas S⁴D uses element-level features for cross-domain clustering and similarity comparison.

2.2 DOM Element Similarity and Tree-Based Comparison

Measuring how similar DOM elements or entire DOM trees are—without relying on rendered images—has attracted a growing body of work.

Grigera et al. [8] proposed two algorithms for the **flexible detection of similar DOM elements**, measuring similarity through a mixed approach that combines an element’s location in the DOM tree with its inner structure. The approach does not require CSS attributes or visual screenshots and was evaluated against a manually annotated ground truth of over 1,200 DOM elements, yielding measurable precision across heterogeneous real-world sites. Their work demonstrates that DOM tree position and sub-tree topology already carry enough discriminative power to identify structurally equivalent elements across pages. S⁴D extends this idea by adding the CSS attribute dimension—treating CSS values as a feature vector alongside structural position—which enables finer-grained distinctions between elements that share topology but differ in visual style.

Brisset et al. [2] proposed **SFTM** (Similarity-based Flexible Tree Matching), a fast algorithm for matching entire web page DOM trees based on node labels and local topology similarity rather than CSS attributes or visual features. SFTM avoids the combinatorial explosion of exhaustive subtree comparison and achieves an average matching time of approximately 182 ms on real-world web documents. The work is notable for demonstrating that practical tree matching at the scale of real web pages is feasible within a structural-only framework. Where SFTM operates at the full-page tree level to determine a single similarity score between two pages, S⁴D operates at the element level, clustering individual CSS-annotated nodes and then aggregating cluster memberships into domain-level role vectors for cross-site comparison.

Beyond pairwise element matching, the extraction of structured information from web pages has been another central research theme, with work from Gulhane et al. [9] on **Vertex**, a wrapper induction system deployed at Yahoo! for large-scale record extraction. Vertex introduced algorithms for grouping similar pages, generating XPath-based extraction rules, and adapting to structural changes—extracting over 250 million records from more than 200 websites. While Vertex demonstrates the scalability of web information extraction and implicitly relies on structural regularity to identify common DOM patterns,

it is tailored to structured record retrieval. S⁴D does not generate wrappers or extract database-like content; it focuses instead on comparing visual and structural layout signatures across entire domains, making these two systems complementary in scope.

2.3 Web Template and Design Pattern Detection

Detecting recurring design patterns and templates across web sites represents a problem closely related to the cross-domain similarity goal of S⁴D.

Larroque et al. [15] proposed a method for **web template extraction based on hyperlink graph analysis**. Their approach identifies a small representative set of pages sharing the same template by analyzing the hyperlink structure of a web site, without requiring visual or NLP-based comparison. Template detection is positioned as a prerequisite for downstream tasks including content extraction, block detection, and web page indexing. By exploiting structural redundancy at the graph level, the method can identify common page frames across large crawls. S⁴D addresses a related but orthogonal goal: rather than grouping pages within a site that share a template, it compares the structural and stylistic signatures of elements across different domains to assess cross-domain similarity.

More recently, Lockard et al. [16] proposed **ZeroShotCeres**, which tackles the challenge of zero-shot relation extraction from semi-structured web pages. Their model employs graph neural networks to capture both textual semantics and visual cues such as layout, font size, and color, enabling generalization to previously unseen templates without template-specific training data. ZeroShotCeres highlights the importance of structural and visual cues for generalizable web analysis. While their focus is on entity-level relation extraction, the reliance on layout and visual features as cross-template signals parallels S⁴D’s use of CSS attributes as cross-domain structural fingerprints. S⁴D, however, does not require any labeled training data and does not target relation extraction, operating instead in a fully unsupervised regime for domain-level similarity.

2.4 Web Content Extraction and Main Content Identification

A closely related and long-standing research thread is *web content extraction*: identifying the main, informative portion of a page and discarding boilerplate such as navigation bars, advertisements, and templated furniture. Although this line of work targets a different unit of analysis than S⁴D—extracting a single content region per page rather than comparing structural signatures across domains—many of the underlying features and assumptions (DOM position, tag statistics, CSS-derived visual cues) directly overlap with the feature space S⁴D builds on.

AL-Ghuribi and Alshomrani [1] provide a **comprehensive survey of web content extraction algorithms**, classifying approaches into five families: wrapper-based, template-detection-based, machine-learning-based, visual-cue-based, and HTML-feature/statistics-based methods. Their analysis of experimental results across seventeen prior techniques shows that HTML-feature and statistics-based methods are the most studied and generally best-performing family, and they distill a set of desiderata for an “optimal” extractor: language independence, freedom from structural assumptions, applicability to both single- and multi-body pages, and independence from any specific template or training corpus. These desiderata align closely with the design goals of S⁴D, which likewise avoids template-specific assumptions and operates directly on CSS-annotated DOM features without labeled training data; however, S⁴D’s goal is cross-domain structural comparison rather than isolating a single main-content region within a page.

An early representative of the heuristic family, Gupta and Hilal [10] proposed a **tag-tree and**

heuristics-based approach that parses a page into a tag tree, merges tag trees of multiple pages from the same site based on matching CSS features, and then applies three ranking heuristics—pattern repeating, standard deviation, and sibling-tag counting—to locate the separators that bound the primary content region. Evaluated on 150 web pages, the standard-deviation heuristic achieved the highest rank-1 accuracy (0.78) among the three. Notably, their tag-tree merging step, which groups nodes across pages of the same site when their CSS features match, is conceptually close to the element-level CSS clustering that underlies S⁴D, though their merge is a rule-based, per-site operation rather than a density-based clustering applied across domains.

Joshi and Liu [13], in work carried out at Hewlett-Packard Laboratories, combined **DOM analysis with natural language processing** to jointly extract the main article text and its associated images. Paragraph-bearing DOM subtrees are first identified using a text-size threshold and document order, after which a Named-Entity-Recognition-based cosine similarity measure links each image’s caption to the article body to discriminate genuine article images from advertisements. The reported text-extraction accuracy exceeds 90%, with roughly 83% of informative images correctly identified. Their reliance on block-tag classification (distinguishing layout-affecting “block” elements from purely visual “style” elements) echoes the block/style distinction implicit in S⁴D’s CSS attribute vectors, but their pipeline terminates in a single extracted region per page rather than a similarity comparison across many pages.

Moving to a learning-based formulation, Uțiu and Ionescu [21] cast content extraction as a **binary classification problem over HTML tags**, engineering a feature vector for each tag from its own attributes (depth, sibling position, tag type, class/id length), up to five ancestors, and aggregate statistics over its descendants. Testing logistic regression, linear SVMs, decision trees, random forests, and MLPs on the CleanEval and Dragnet benchmarks, they report a new state-of-the-art F₁ of 0.96 on Dragnet using minimal feature engineering. This is directly relevant to S⁴D in that their node/ancestor/descendant feature scheme demonstrates that DOM-position-derived numerical features alone (without visual rendering) are highly discriminative—supporting S⁴D’s choice to build its similarity pipeline purely from CSS and DOM-position attributes. The key difference is that their features feed a supervised per-tag classifier trained on labeled boilerplate/content annotations, whereas S⁴D’s HDBSCAN stage is entirely unsupervised and operates across, rather than within, pages.

Morbieu et al. [17] proposed a fully **unsupervised three-stage pipeline**: text blocks within a single page are first clustered using a spectral clustering algorithm over a distance defined on weighted DOM tag-paths, the cluster(s) corresponding to the main content are then selected via a stop-word-density heuristic, and finally a classifier (trained on the automatically-produced labels) is used for fast prediction on new pages. On a corpus of company and cultural-institution web sites, their clustering step alone achieved F₁ scores of 79.4 and 81.1, outperforming both Boilerpipe and Newspaper3k. This work is the closest in spirit to S⁴D among the content-extraction literature, since both rely on unsupervised clustering of DOM/CSS-derived block features rather than labeled training data; the crucial distinction is scope and granularity—Morbieu et al. cluster text blocks *within a single page* to separate content from noise, whereas S⁴D clusters CSS-annotated elements *across many pages and domains* to characterize structural similarity, with no notion of a single “main content” cluster to be selected.

Prasath and Ozturk [19] addressed a specialized instance of the problem—extracting content from **scientific articles**—using **Case-Based Reasoning (CBR)**. Their system classifies HTML tag sequences as informative or noisy by first checking a growing case base of previously seen sequences and, for genuinely new sequences, applying a count-and-level heuristic that favors tag sequences recurring frequently and overlapping at multiple tree depths (consistent with how repeated article sections are marked up). Evaluated on articles from six publishers (Springer, Wiley, Macmillan), the approach achieved purity scores between 85% and 96% and accuracy between 88% and 95%. While narrower in domain than S⁴D, the count-and-level assumption—that structurally recurring, similarly-nested tag sequences signal a coherent

content type—parallels S⁴D’s premise that recurring CSS-annotated DOM patterns are informative of a page’s or domain’s structural role.

Singh [20], in a Master’s thesis combining **image processing with markup analysis**, first renders a web page to an image and applies edge-detection-based segmentation to identify visually distinct regions (header, footer, navigation, body, etc.), then maps the segmented regions back onto the DOM tree and classifies each region’s role using either a rule-based or an SVM-based approach over spatial and markup features (text length, hyperlink density, font size and weight). On a corpus of 150 pages (1,222 regions), the segmentation step achieved 77.8% overall accuracy, with role classification reaching 74–75% for either approach individually and 81.3% when the two are combined. This thesis is notable for explicitly motivating region and role identification as a precursor to *web-page similarity assessment*—the same high-level goal as S⁴D—but it pursues this goal via rendered-image segmentation combined with DOM mapping for a single page’s layout, whereas S⁴D never renders pages and instead compares raw CSS feature vectors of DOM elements directly across domains.

Jung et al. [14] tackled the problem of **multilingual main content extraction using visual features**, observing that prior methods such as Readability.js, DOM Distiller, Web2Text, and BoilerNet lose up to 40% performance on non-English pages because they rely on language-dependent textual cues (word counts, punctuation, stop words). Their grid-centering-expanding (GCE) method instead defines three candidate centroids—derived from the browser window, the full document, and a proposed “first browsing area”—finds the DOM leaf node nearest each centroid, and expands upward through ancestors using language-independent rules (the presence of an `<article>` tag, positive attribute keywords, and bounded width growth) until a main-content candidate is found. Across nine datasets in eight languages, GCE improved the Longest Common Subsequence F₁ score by 9–21% and the text-block-matching F₁ score by 16–43% over existing methods. GCE’s use of positional/geometric features (distance from computed centroids, element width and height) mirrors the CSS-derived positional and size attributes exploited by S⁴D, reinforcing the idea that layout geometry, independent of textual language, is a strong structural signal; however, GCE operates per-page to isolate one content region, while S⁴D aggregates CSS-annotated element clusters into domain-level role vectors for cross-domain comparison.

Most recently, and reflecting a shift in the motivation for content extraction toward Large Language Model (LLM) pipelines, Chen et al. [5] introduced **Index-based Web Content Extraction**, which reframes extraction as a discriminative index-prediction task rather than a generative one. HTML is first segmented into DOM-aligned, addressable blocks; a fine-tuned model (“IndexLM”, built on Qwen3 at 0.6B–4B parameters) then predicts the indices of blocks relevant to a query (or, when no query is given, to main-content extraction) instead of regenerating the content token by token. On both a retrieval-augmented-generation question-answering benchmark and a direct main-content-extraction benchmark, IndexLM outperformed generative baselines (ReaderLM-v2, prompted LLMs, HtmlRAG) in F₁ score while being up to an order of magnitude faster. Although this work is motivated by LLM context-window management rather than structural similarity analysis, its DOM-aligned block segmentation is structurally identical to the DOM-element granularity S⁴D operates on, illustrating that fine-grained, structure-aware block decomposition of HTML remains a foundational step across otherwise very different downstream tasks—extractive summarization for LLMs on one hand, and unsupervised structural similarity clustering on the other.

2.5 Density-Based Clustering and Clustering Validation

The clustering layer of S⁴D relies directly on HDBSCAN and the DBCV validity index. Both have been formally introduced and analyzed in the literature.

The foundational density-based algorithm DBSCAN [7] groups points by neighborhood density but

requires a global epsilon threshold, making it sensitive to clusters of varying density. Campello et al. [3] introduced **HDBSCAN** (Hierarchical Density-Based Spatial Clustering of Applications with Noise), a density-based clustering algorithm that constructs a complete hierarchy of clusters across all possible density thresholds following Hartigan’s density-contour model. From this hierarchy, HDBSCAN extracts a flat partitioning in a fully unsupervised manner by maximizing cluster *stability*—selecting clusters that persist over a wider density range (i.e., whose stability exceeds the sum of their children’s stabilities)—without requiring the user to specify an epsilon threshold as in traditional DBSCAN. In addition, HDBSCAN assigns each point a GLOSH outlier score in $[0, 1]$ that unifies global and local outlier perspectives. These properties make HDBSCAN well-suited to high-dimensional, heterogeneous web data where clusters vary widely in density and shape, precisely the setting of S^4D ’s CSS feature space.

Moulavi et al. [18] introduced **DBCV** (Density-Based Clustering Validation), a clustering validity index designed specifically for density-based algorithms. The motivation behind DBCV is that traditional validity indices—including Silhouette, Davies-Bouldin, and Calinski-Harabasz—assume globular or spherical cluster geometry and are therefore inappropriate for evaluating the output of algorithms like HDBSCAN, which can discover arbitrarily shaped clusters. DBCV provides a principled alternative that respects the density-based structure of the solution. S^4D adopts DBCV as its cluster quality metric to guide hyperparameter selection (notably the minimum cluster size) and to compare alternative CSS feature configurations, ensuring that the chosen clustering reflects genuine density structure in the element-level feature space rather than artifacts of spherical assumptions.

2.6 Comparison with Prior Work

The reviewed works collectively advance web structural analysis along multiple complementary dimensions, yet none directly targets the specific combination of concerns addressed by S^4D :

- **WebSRC** [4]: Focuses on question answering over structured web data, emphasizing the need for DOM-aware structural comprehension. S^4D uses structural encoding not for query answering but for unsupervised cross-domain similarity analysis.
- **DOM-LM** [6]: Learns generalizable document embeddings from DOM and text jointly, requiring fine-tuned transformer training. S^4D operates on raw CSS feature vectors with no training phase, applying unsupervised clustering directly.
- **Hambley et al.** [11]: Demonstrate empirically that DOM structural features outperform content features for page clustering, applied to accessibility evaluation within single sites. S^4D extends this structural clustering paradigm to the element level and aggregates results for cross-domain comparison.
- **Huynh et al.** [12]: Perform intra-page segmentation via DOM-structural cohesion analysis, using element types and attributes as features. S^4D shares the element-level feature perspective but applies it to inter-domain clustering rather than intra-page boundary detection.
- **Grigera et al.** [8]: Measure DOM element similarity via tree position and inner structure. S^4D augments this structural signal with CSS attribute values, enabling style-sensitive similarity alongside topological similarity.
- **Brisset et al. (SFTM)** [2]: Match entire page DOM trees efficiently using node labels and local topology. S^4D operates at the element level rather than the full-page tree level, clustering fine-grained CSS-annotated nodes before aggregating to domain-level role vectors.

- **Vertex** [9]: Extracts structured records at web scale via wrapper induction. S⁴D does not extract records but compares layout and style fingerprints across domains.
- **ZeroShotCeres** [16]: Performs zero-shot relation extraction using layout and visual features modeled by GNNs. S⁴D uses similar structural cues but requires no labeled data and does not target entity-level relations.
- **Larroque et al.** [15]: Detects intra-site template groups via hyperlink graph analysis. S⁴D targets cross-domain similarity rather than within-site template clustering.
- **AL-Ghuribi and Alshomrani** [1]: Survey classifying web content extraction methods into five families and distilling desiderata for a template- and language-independent extractor. S⁴D satisfies these same desiderata but applies them to cross-domain structural comparison rather than single-page content isolation.
- **Gupta and Hilal** [10]: Merge tag trees across pages of a site based on matching CSS features, then apply ranking heuristics to locate content separators. S⁴D generalizes CSS-based node merging from a rule-based, per-site heuristic into density-based clustering applied across domains.
- **Joshi and Liu** [13]: Combine DOM block/style tag classification with NLP-based image-caption similarity to extract a single article region. S⁴D shares the block-level DOM feature perspective but clusters elements across many pages rather than isolating one region per page.
- **Uȕiu and Ionescu** [21]: Classify individual HTML tags as content/boilerplate using supervised models over node, ancestor, and descendant DOM features. S⁴D uses a structurally similar DOM-derived feature philosophy but clusters unsupervised across domains rather than classifying within a single page.
- **Morbieu et al.** [17]: Unsupervised three-stage pipeline that clusters DOM tag-path features within a page to isolate the main content cluster. S⁴D is the closest in spirit but clusters CSS-annotated elements across pages and domains rather than within one page.
- **Prasath and Ozturk** [19]: Case-Based Reasoning over recurring, level-overlapping tag sequences to extract scientific-article content. S⁴D generalizes the underlying premise—that recurring structural patterns are informative—beyond a single domain (scientific publishers) to arbitrary web domains.
- **Singh** [20]: Segments a rendered page image into regions and classifies their roles via markup features, explicitly motivated by web-page similarity assessment. S⁴D pursues the same similarity goal without rendering pages, operating directly on CSS feature vectors of DOM elements.
- **Jung et al. (GCE)** [14]: Uses language-independent positional/geometric DOM features to extract main content across multilingual pages. S⁴D likewise relies on layout-geometry features (via CSS) rather than textual language, but aggregates them into domain-level role vectors instead of isolating one region per page.
- **Chen et al. (IndexLM)** [5]: Reframes LLM-oriented content extraction as index prediction over DOM-aligned HTML blocks. S⁴D operates at the same DOM-block granularity but for unsupervised structural similarity rather than LLM context management.
- **HDBSCAN** [3]: The foundational density-based clustering algorithm adopted by S⁴D. The paper establishes the stability-maximization extraction strategy that S⁴D directly exploits for element-level CSS clustering.

- **DBCV [18]:** The validity index used by S⁴D to evaluate and tune its HDBSCAN configurations, selected because it handles the non-globular cluster shapes that arise in CSS feature spaces.

2.7 Research Gaps and Opportunities

The literature reveals several gaps that motivate the design of S⁴D:

- **Cross-domain structural similarity:** The majority of reviewed systems operate within a single website or across pages sharing a known template. None compute a similarity matrix across structurally heterogeneous domains, which is the primary output of S⁴D.
- **Element-level CSS feature clustering:** Existing work on DOM element similarity (Grigera et al., Huynh et al.) relies on tag types and tree topology. Incorporating the CSS attribute dimension—font sizes, margins, padding, display properties—as a continuous feature vector for clustering is largely unexplored.
- **Multi-dimensional aggregation:** Prior clustering-based web analysis (Hambley et al.) clusters entire pages. S⁴D introduces a two-level approach: clustering at the element level, then aggregating cluster membership distributions into domain-level role vectors for inter-domain comparison.
- **Unsupervised, label-free pipeline:** Systems like DOM-LM and ZeroShotCeres require either labeled data or fine-tuning. S⁴D achieves cross-domain similarity estimation in a fully unsupervised manner using only raw CSS attributes and the HDBSCAN/DBCV stack.
- **Scalable cross-domain analysis:** While Vertex demonstrates extraction scale, scalable structural similarity across diverse domains remains underexplored. S⁴D’s pipeline is designed to process multiple domains concurrently and aggregate results at the domain level.
- **Cluster quality evaluation for web features:** Template and tree-matching approaches [15, 2] do not evaluate cluster cohesion. S⁴D addresses this through DBCV [18], paired with HDBSCAN [3], providing a principled quality metric for the non-globular CSS feature clusters that arise in practice.
- **Single-region extraction versus cross-page structural clustering:** The web content extraction literature [1, 10, 13, 21, 17, 19, 20, 14, 5] is uniformly framed around isolating a single main-content region *within one page*, whether via heuristics, supervised classification, unsupervised clustering, CBR, image segmentation, visual geometry, or LLM-based index prediction. None of these methods cluster DOM/CSS-annotated elements *across* pages or domains; S⁴D repurposes the same class of DOM/CSS features for this fundamentally different, cross-domain clustering objective.
- **Language- and rendering-independent structural features:** Several content-extraction methods [14, 20] demonstrate that visual/geometric or rendered-image features are effective but require a rendering or screenshot step. S⁴D achieves a comparable structural signal using only CSS attributes extracted from the DOM, avoiding the computational cost of page rendering while remaining language-independent.

2.8 Summary

This chapter has reviewed twenty representative systems and methods that collectively span the major technical dimensions underlying S⁴D: structural web understanding (WebSRC, DOM-LM, Hambley et al., Huynh et al.), DOM element and tree-based similarity (Grigera et al., Brisset et al./SFTM, Vertex),

web template and pattern detection (Larroque et al., ZeroShotCeres), web content extraction and main content identification (AL-Ghuribi and Alshomrani, Gupta and Hilal, Joshi and Liu, Uȕiu and Ionescu, Morbieu et al., Prasath and Ozturk, Singh, Jung et al., Chen et al.), and density-based clustering with validation (HDBSCAN by Campello et al., DBCV by Moulavi et al.). Each system contributes important ideas relevant to S⁴D, yet none directly combines element-level CSS feature extraction, density-based clustering with DBCV-guided tuning, and cross-domain role-vector aggregation into a single unsupervised pipeline. In particular, the web content extraction literature confirms that DOM position, tag statistics, and CSS-derived geometric features are strong, language-independent signals for structural analysis, but consistently applies them to isolate a single content region *within* a page rather than to compare structural signatures *across* pages and domains. The gaps identified above motivate the design choices presented in the following chapter.

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